APEX RETAIL GROUP | INDEPENDENT HOMEWORK ASSIGNMENT

DOCUMENT: CLASS HOMEWORK (SESSION 7)

SCENARIO: SUPPLY CHAIN VENDOR SHIPMENT & WAREHOUSE INVENTORY MERGE

ASSIGNMENT OVERVIEW: You are assisting the Supply Chain Director at Apex Retail Group. The director suspects that delivery delays are occurring because specific regional warehouses are receiving shipments from high-latency logistics vendors.

To prove this, you must merge vendor shipment logs with warehouse destination logs, calculate delivery turnaround times, and identify bottleneck origins.

RAW DATA TABLES:

```python
import pandas as pd

shipment_logs = {
    "shipment_id": ["SHP-11", "SHP-12", "SHP-13", "SHP-14", "SHP-15", "SHP-16", "SHP-17", "SHP-18"],
    "vendor_id": ["VEND-A", "VEND-B", "VEND-A", "VEND-C", "VEND-B", "VEND-C", "VEND-A", "VEND-B"],
    "transit_days": [3.5, 7.2, 4.0, 2.1, 8.5, 2.5, 3.8, 6.9],
    "freight_cost": [400.0, 1200.0, 450.0, 210.0, 1350.0, 230.0, 420.0, 1100.0]
}

warehouse_assignments = {
    "shipment_id": ["SHP-11", "SHP-12", "SHP-13", "SHP-14", "SHP-15", "SHP-16", "SHP-17"], # SHP-18 missing assignment
    "destination_warehouse": ["North-DC", "South-DC", "North-DC", "East-DC", "South-DC", "East-DC", "North-DC"],
    "manager_on_duty": ["Sarah", "David", "Sarah", "Michael", "David", "Michael", "Sarah"]
}
```

PROBLEM 1: RELATIONAL MERGING & DATA RETENTION AUDIT

In a Jupyter Notebook named `homework_session7_yourname.ipynb`:

1. Load `shipment_logs` and `warehouse_assignments` into DataFrames `df_shipments` and `df_warehouses`.
2. Perform a **Left Join** (`how='left'`) keeping all records from `df_shipments` as the left table and merging with `df_warehouses` on `"shipment_id"`.
3. Print the resulting DataFrame shape and check for any `NaN` values in warehouse columns (identifying unassigned shipment `SHP-18`).

PROBLEM 2: GROUPED AGGREGATION BY VENDOR AND WAREHOUSE

1. Fill missing warehouse assignments with `"Unassigned"` using `.fillna()`.
2. Group the merged DataFrame by `vendor_id` and calculate average `transit_days` and total `freight_cost`.
3. Group the merged DataFrame by `destination_warehouse` and calculate average `transit_days`.
4. Print both summary tables.

PROBLEM 3: SUPPLY CHAIN BOTTLENECK REPORT

Write a 150-word formal analytical report for the Supply Chain Director containing:

1. Identification of the vendor with the highest average transit days and freight cost.
2. Identification of the destination warehouse experiencing the longest average delivery turnaround.
3. Strategic recommendations on whether to terminate vendor contracts or adjust warehouse logistics routing.

SUBMISSION DEADLINE: Prior to Session 8 (Mid-Course Checkpoint).